



Quantum-Enhanced Adaptive Urban Traffic Optimization

Problem Statement

Urban traffic congestion is a major challenge in growing cities. Traditional traffic signals often use fixed timings and cannot efficiently respond to changing traffic density, accidents, road closures, or emergency vehicles. Poor coordination between nearby intersections can also increase waiting time, fuel consumption, and carbon emissions.

The proposed system aims to develop a **Hybrid Quantum-Classical Traffic Optimization Platform** that dynamically manages traffic signals across multiple interconnected intersections.

The system should model traffic flow as an optimization problem and explore techniques such as **QUBO and QAOA** to determine efficient signal timings while considering vehicle density, queue length, road capacity, pedestrian movement, and emergency vehicle priority.

A major feature should be an **Emergency Green Corridor System** that reduces ambulance or emergency vehicle travel time while minimizing disruption to normal traffic.

Core Features

- **Multi-Intersection Traffic Network**

Model approximately **4-8 connected intersections** with traffic density, queue length, road capacity, and signal status.

- **Quantum Optimization Engine**

Use **QUBO/Ising formulation and QAOA** or a hybrid quantum-classical approach to optimize signal timings.

- **Adaptive Traffic Signals**

Automatically adjust green-light duration based on changing traffic conditions instead of using only fixed timings.

- **Emergency Green Corridor**

Prioritize ambulances or emergency vehicles by dynamically modifying selected traffic signals and restoring normal traffic afterward.

• **Dynamic Event Handling**

The system should respond to at least one event such as:

- Sudden traffic congestion
- Accident
- Road closure
- Emergency vehicle arrival

• **Environmental Analysis**

Estimate improvements in:

- Vehicle waiting time
- Traffic throughput
- Fuel consumption
- CO₂ emissions

• **Classical Comparison**

Compare the hybrid quantum solution with a basic method such as fixed signal timing or rule-based traffic control.

• **Interactive Dashboard**

The dashboard should display:

- Road network
- Traffic density
- Current and optimized signals
- Emergency vehicle route
- Queue length
- Waiting time
- Fuel and CO₂ estimates
- Classical vs quantum results

Optimization Objectives

The system should aim to:

- Minimize waiting time
- Minimize queue length
- Reduce congestion
- Reduce emergency travel time
- Reduce fuel usage and emissions
- Maximize traffic throughput

Suggested Development Stack

The project should be buildable using free tools such as:

- Qiskit / Qiskit Aer
- PennyLane
- Python
- NetworkX
- SUMO or custom traffic simulation
- Streamlit
- Folium / OpenStreetMap

Expected Outcome

A working prototype that demonstrates how **hybrid quantum-classical optimization** can improve multi-intersection traffic flow, reduce congestion and emissions, and provide priority routing for emergency vehicles.

The solution should clearly explain where the quantum component is used and compare its results with a simple classical baseline.